Report Nº 1
Team Out of Lane
Date 15.12.2024

1 Introduction

This report details the activities carried out to date in this first sprint. These activities were focused on:

1. Designing the foundations of our solution to the challenge (architecture, hardware needs, basic team organization).
2. Understanding the provided code and hardware as it is.

We understand this strategy will enable us to create a robust framework for future development.

2 Planned activities

Category	*Activity*	*Responsible*
Hardware improvements	Mechanical calibration	Alexander P.
	Research GPU addition	Alexander P. and Evelyn C.
	Research hardware additions (ultrasonic sensors and headlight)	Alexander P. and Evelyn C.
Sensing and input package	Camera handling, ROIs definition	The entire team.
Computer vision package (RPi)	Research lane detection	Alexander P. Carolina N. and Pablo F.
	Research YOLO for object detection	Pablo F. and Victor P.
Project Documentation	Set up development environment	The entire team.
	Embedded system components diagram	Alexander P. and Evelyn C.
	Project plan	Carolina N. and Pablo F.
	Architecture diagram	Pablo F.
	Document the 1st report (robot can be controlled remotely)	Carolina N. and Pablo F.

3 Status of planned activities

Activity	*% completed*	*Observations*
Mechanical calibration	50%	The basic setup is complete. However, there is still a slight deviation in the car's direction.
Research GPU addition	75%	We are still trying to acquire a GPU that fits our use cases and available resources. So far, out main options are: ▪ Jetson Orin Nano ▪ RPi AI Kit
Research hardware additions (ultrasonic sensors and headlight)	50%	We are still researching and discussing this topic, though ultrasonic sensors will be certainly acquired.
Camera handling, ROIs definition	100%	We have defined a ROI for the camera in the Gazebo simulator and in the real car.
Research lane detection	100%	We will develop a lane detection module in the brain (as showcased in the architecture diagram) based on OpenCV.
Research YOLO for object detection	100%	We will develop an object detection module in the brain based on YOLO v11.
Set up development environment	100%	We set up the brain, the rpi-imager, the Gazebo simulator, and a conda environment for local testing of our computer vision algorithms.

Activity	*% completed*	*Observations*
Embedded system components diagram	100%	This diagram will help us understand the embedded system and how we might change it (see "Research hardware additions").
Project plan	100%	We agreed on a tentative project plan for the team.
Architecture diagram	100%	One of the deliverables of this first report.
Document the 1st report (robot can be controlled remotely)	100%	The report itself.

4 General status of project

The team has been successful in the completion of fundamental tasks focused on architecture and team workflow. On the other hand, we have been slightly delayed on tasks related to the hardware improvements and calibration.

The car can be moved manually using the web application. It moves forward, backward, and can perform maneuvers such as turning to both sides. It receives data from the camera correctly and can stream video to the dashboard.

5 Upcoming activities

To be continued from the 1st sprint

- **Mechanical calibration**
- **Research GPU addition**
- **Research hardware additions (ultrasonic sensors and headlight)**

New activities

- **Implement Lane Detection**
  We should make the robot capable of detecting the lane.
- **Implement traffic sign detection**
  We should implement a basic traffic signals detection model based on an inference model, such as YOLO.
- **Implement basic controller**
  We should make the robot capable of following simple instructions, such as "go ahead following the lane".
- **Architecture revision**
  We should revisit the architecture definition to check if it needs to be adjusted.
- **Preparation of traffic signals dataset**
  We should prepare an application-specific image dataset in order to train a better suited AI model for traffic signals detection.